For example, the following patterns such as,

$$\begin{aligned} 7 + 9 &= 9 + 7 \\ 57 + 43 &= 43 + 57 \\ 123 + 456 &= 456 + 123 \\ 7098 + 2018 &= 2018 + 7098 \\ 35784 + 481269841 &= 481269841 + 35784 \end{aligned}$$

can be simply summarized as $a + b = b + a$.

Here, we have two variables namely *"a"* and *"b"*. Each variable can take *"any value"*, but the value of *'a'* is the same on both sides, and the value of *'b'* is also the same on both sides. But, the values of *'a'* and *'b'* need not be equal to each other.

Can you give a similar intrepretation for $a \times b = b \times a$?

NOTE

Similar rules cannot be written for subtraction! We know what is '7 – 3', but not '3 – 7'. So *'a – b'* and *'b – a'* are not the same!

2.3 Understanding operations on Variables

Consider the following situations.

Situation 1:



Mathi is 3 years elder than his sister Nila. If we know Nila's age, can we find Mathi's?

If Nila's age is *'n'*, you can see that Mathi's age is always *'n + 3'*. This is the advantage of using variables. We do not need different statements for different values of the age! As we give different values for *'n'*, we get different values for *'n + 3'*, here 3 is a *constant*. This is clear from the following table.

Nila's age *'n'*	Mathi's age 'n + 3'
If n = 4	7
If n = 8	11
If n = 12	15

6th_Chapter 2_032-045.indd 35 1/27/2020 1:42:55 PM